Enterprise AI Customer Intelligence Platform

Solution Architecture & Implementation Guide

Enterprise Analytics Platform

Microsoft Fabric

Data Engineering • Artificial Intelligence • Business Intelligence

OneLake • Lakehouse • Warehouse • PySpark • Azure AI Foundry • GPT-5 • Power BI

Version 1.0

July 2026

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Solution Overview

Key Takeaway

The Enterprise AI Customer Intelligence Platform demonstrates how Microsoft Fabric can be used to build an end-to-end enterprise analytics solution that combines Data Engineering, Artificial Intelligence and Business Intelligence within a modern Medallion Architecture.

Business Challenge

Organizations generate customer data from multiple systems including sales, products and customer reviews. While these datasets provide valuable business insights, they are often fragmented and customer feedback remains unstructured, limiting its analytical value.

Solution

The platform centralizes enterprise data within Microsoft Fabric using a Medallion Architecture. Customer reviews are enriched with Azure AI Foundry (GPT-5), transformed into structured business intelligence and loaded into a Fabric Warehouse using a Galaxy Schema. Power BI delivers interactive dashboards through a governed Semantic Model.

Technology Stack

Capability	Technology
Data Engineering	Fabric Data Factory, OneLake, Lakehouse, PySpark
Artificial Intelligence	Azure AI Foundry (GPT-5)
Analytics	Fabric Warehouse, Power BI Semantic Model
Version Control	GitHub

Business Value

- Unified Customer 360 analytics
- AI-powered customer intelligence
- Enterprise-grade reporting
- Governed semantic model
- Scalable Microsoft Fabric architecture

Solution Architecture

Key Takeaway

The solution follows a layered enterprise architecture that separates data ingestion, engineering, AI enrichment, analytical storage and business reporting into independent services.

The platform integrates Microsoft Fabric services into a single analytical workflow. Data is ingested using Fabric Data Factory, stored within OneLake, transformed through the Medallion Architecture, enriched using Azure AI Foundry and loaded into a Fabric Warehouse before being consumed through a Power BI Semantic Model.

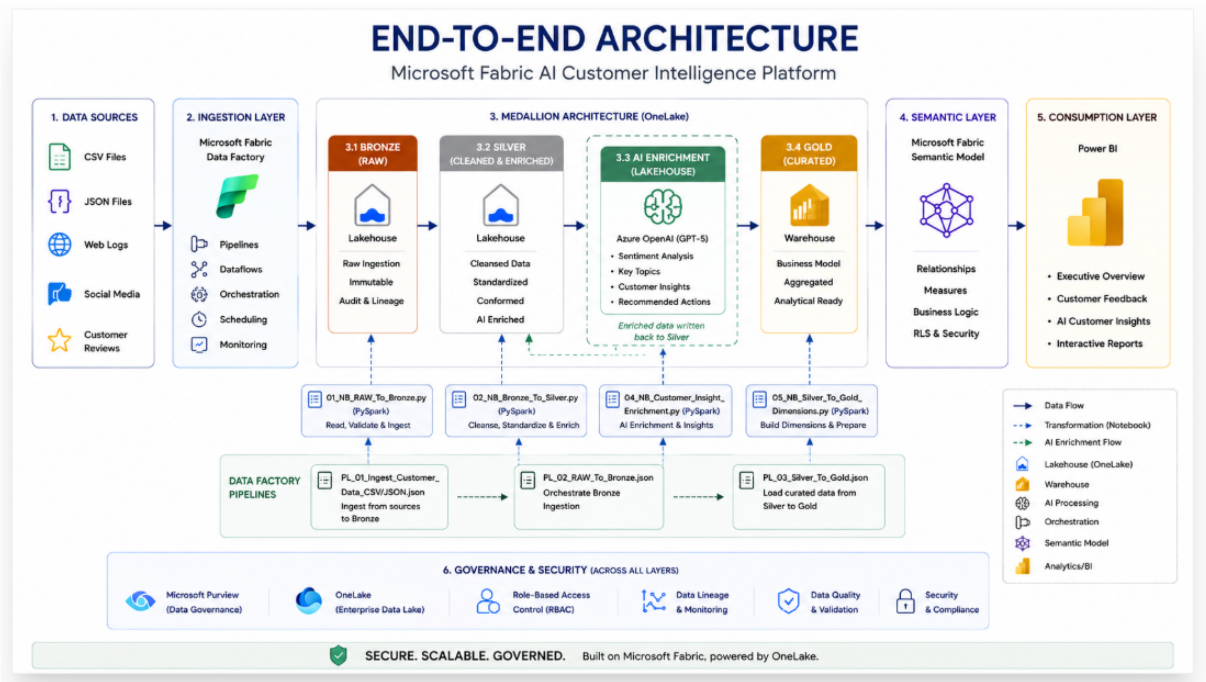

Figure 2.1. End-to-End Enterprise Architecture

Architecture Components

Component	Responsibility
Data Factory	Data ingestion
OneLake	Centralized storage
Lakehouse	Bronze and Silver layers
Azure AI Foundry	AI enrichment
Warehouse	Gold analytical model
Power BI	Business reporting

Medallion Architecture

Key Takeaway

The Medallion Architecture progressively improves data quality by separating raw ingestion, business transformation, AI enrichment and analytical modelling into dedicated processing layers.

The platform adopts Microsoft's recommended Medallion Architecture to transform raw operational data into trusted analytical datasets.

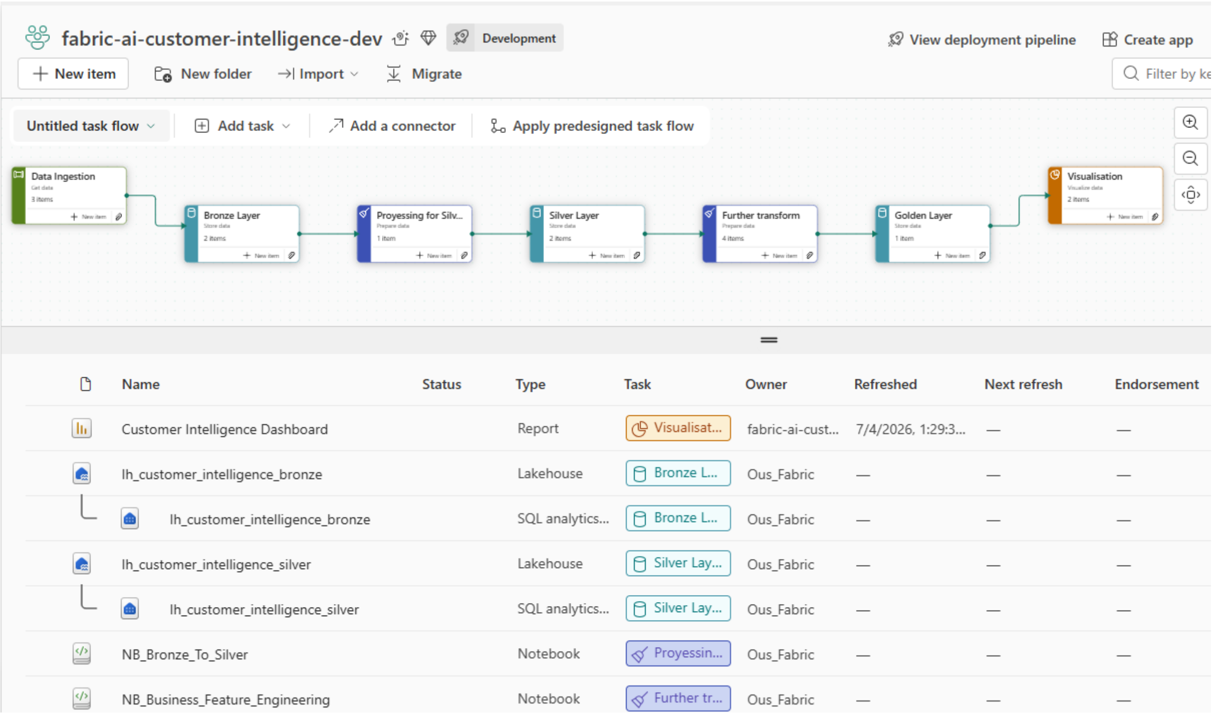

Figure 3.1. Microsoft Fabric Medallion Architecture

Processing Layers

Layer	Purpose
Bronze	Raw enterprise data
Silver	Cleansing and feature engineering
AI	Customer review enrichment
Gold	Business-ready analytical model

The separation of responsibilities simplifies governance, improves data quality and enables reusable analytical datasets.

AI Enrichment

Key Takeaway

Artificial Intelligence is integrated directly into the Silver layer, transforming unstructured customer reviews into trusted business intelligence before loading the Gold analytical layer. By enriching data once during data engineering, AI-generated insights become reusable enterprise assets for all downstream reporting and analytics.

Customer reviews are processed using Azure AI Foundry (GPT-5) after data cleansing and feature engineering. Rather than executing AI at the reporting layer, the enrichment process generates structured business attributes that are persisted within the Lakehouse and promoted to the Fabric Warehouse.

AI Enrichment Workflow

Customer Reviews

↓

Azure AI Foundry (GPT-5)

↓

Sentiment • Category • Priority • Recommendation

↓

Gold Warehouse

↓

Power BI Semantic Model

Generated Business Attributes

Attribute	Purpose
Sentiment	Customer satisfaction analysis
Category	Complaint classification
Priority	Business impact assessment
Summary	AI-generated review summary
Keywords	Trend and topic analysis
Recommended Action	Operational recommendations

Example AI Output


```
{
  "sentiment": "Negative",
  "category": "Delivery",
  "priority": "High",
  "summary": "Delayed delivery and damaged package.",
  "keywords": [
      "delivery",
      "shipping",
      "damage"
  ],
  "recommended_action":
      "Escalate to Logistics Team"
}
```


Architectural Decision

AI enrichment is performed within the Silver layer rather than during reporting. This approach creates reusable enterprise data assets, ensures consistent business definitions, eliminates repeated AI processing and improves Power BI report performance.

Data Warehouse & Semantic Model

Key Takeaway

A Galaxy Schema combined with a Power BI Semantic Model enables multiple business domains to share common dimensions while maintaining independent fact tables.

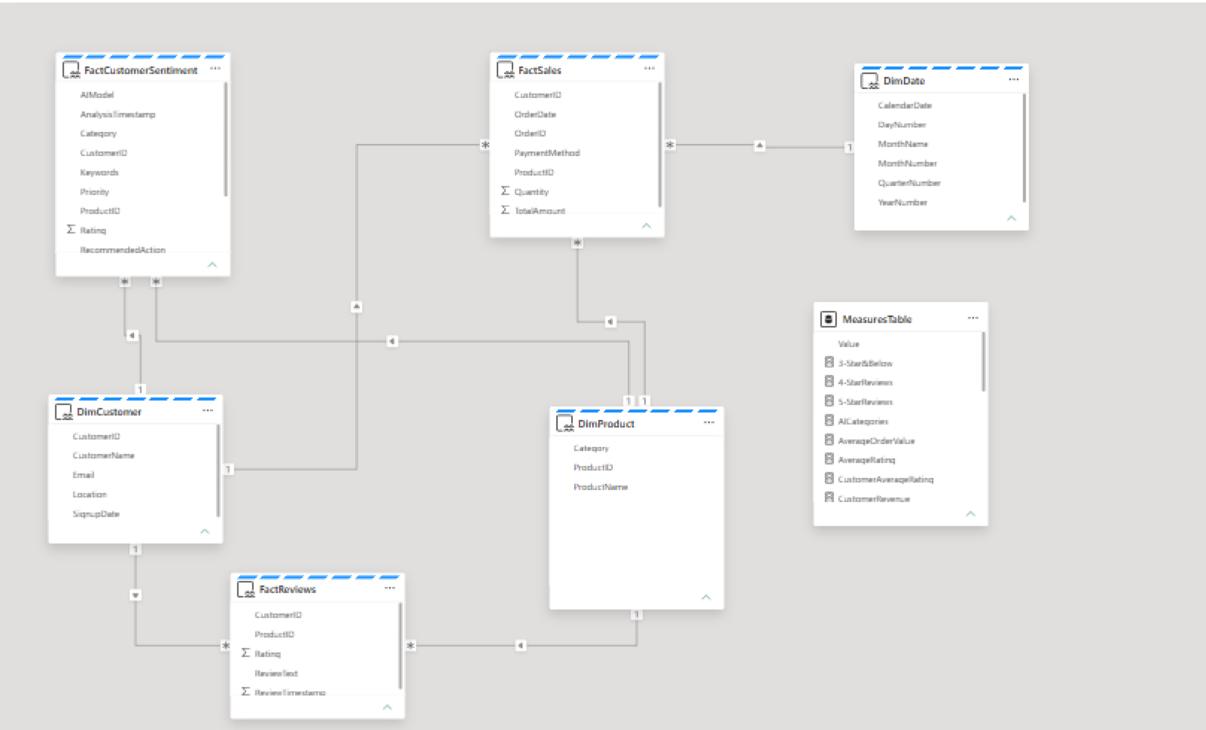

Figure 5.1. Enterprise Galaxy Schema

Analytical Objects

Object	Purpose
DimCustomer	Customer analytics
DimProduct	Product analytics
DimDate	Time intelligence
FactSales	Sales metrics
FactReviews	Review metrics
FactCustomerSentiment	AI insights
Measures Table	Centralized KPIs

Business Intelligence

Key Takeaway

Interactive Power BI dashboards transform curated analytical datasets into actionable business intelligence, enabling executives and operational teams to monitor performance, understand customer behaviour and identify AI-driven insights through a unified semantic model.

The Enterprise AI Customer Intelligence Platform delivers three interactive Power BI dashboards, each designed to answer specific business questions and support data-driven decision making.

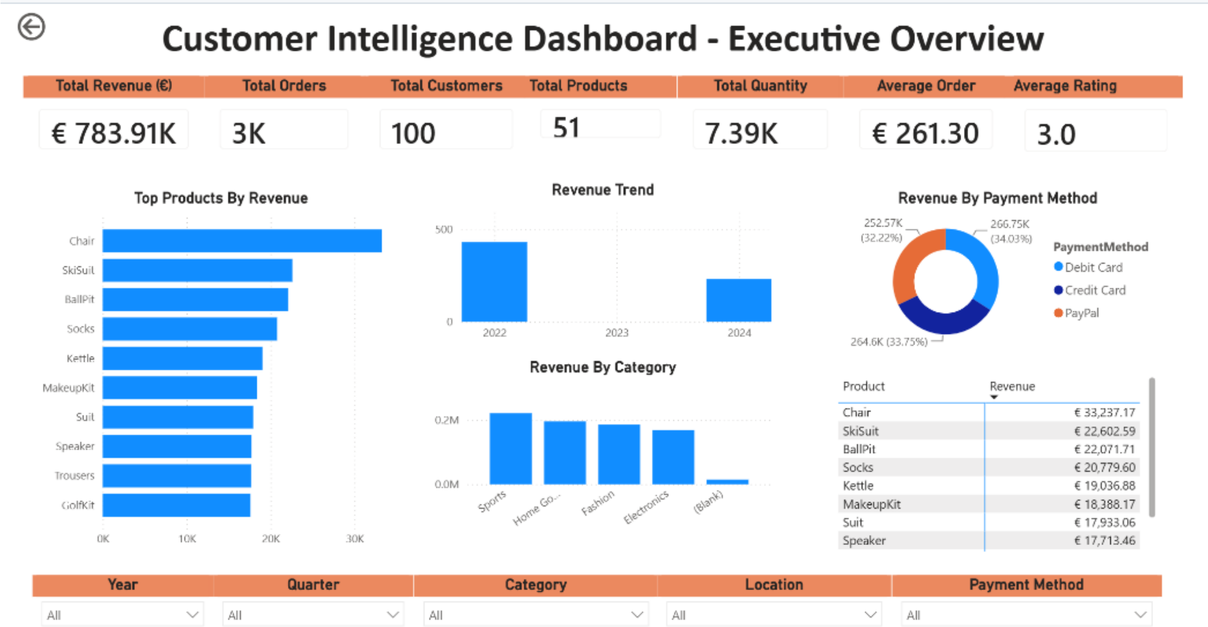

Figure 6.1. Executive Overview Dashboard

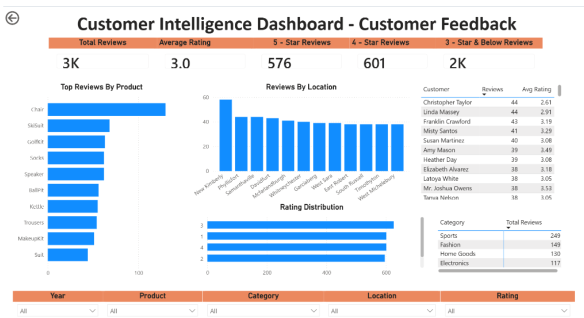

Figure 6.2. Customer Feedback Dashboard

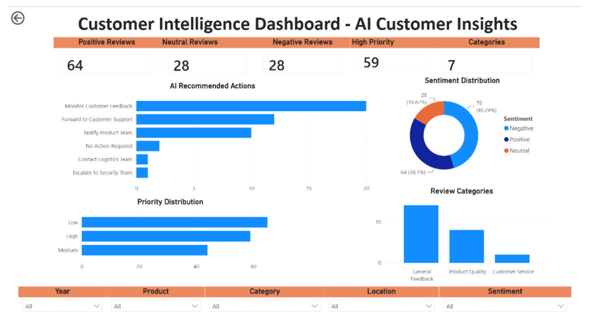

Figure 6.3. AI Customer Insights Dashboard

Dashboard Highlights

- Executive Overview monitors enterprise KPIs including revenue, orders, growth and customer satisfaction.
- Customer Feedback provides insight into customer ratings, review activity and product performance.
- AI Customer Insights leverages GPT-5 enriched data to identify customer sentiment, complaint categories, business priorities and recommended actions.

Business Questions Answered

Dashboard	Business Questions Answered
Executive Overview	Revenue performance, sales trends, customer satisfaction, top-performing products and executive KPIs.
Customer Feedback	Customer ratings, review distribution, product feedback, review trends and customer engagement.
AI Customer Insights	Customer sentiment, complaint categories, business priority, recommended actions and AI-generated insights.

Deployment Strategy

Key Takeaway

Microsoft Fabric Deployment Pipelines provide controlled promotion of analytical assets across Development, Test and Production environments.

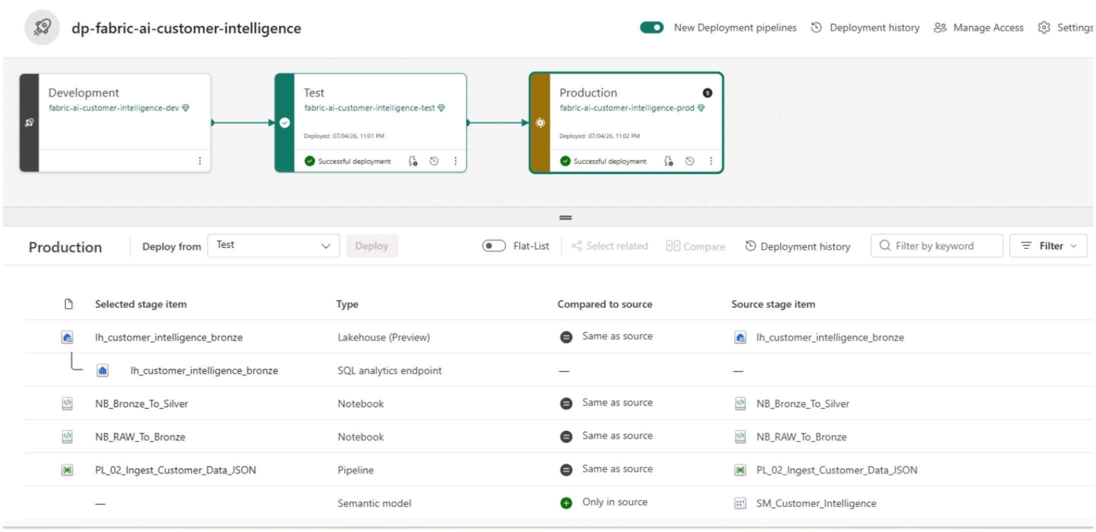

Figure 7.1. Deployment Pipeline

Deployment Roadmap

Current Capability	Future Enhancement
GitHub Repository	Fabric Git Integration
Azure DevOps Repository	Automated CI/CD
Deployment Pipelines	GitHub Actions
Enterprise Workspace	Production Monitoring

Conclusion

Key Takeaway

The Enterprise AI Customer Intelligence Platform demonstrates how Microsoft Fabric can be used to build scalable, AI-powered analytical solutions using modern enterprise architecture principles.

Key Achievements

- End-to-end Microsoft Fabric implementation
- AI-powered customer intelligence
- Enterprise Galaxy Schema
- Power BI Semantic Model
- Executive dashboards
- Deployment Pipelines

Future Enhancements

- Streaming ingestion
- Incremental refresh
- Fabric Git Integration
- Automated CI/CD
- RAG-powered customer support

Thank You

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